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Benchmarking regression algorithms for loss given default modeling

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ABSTRACT

The introduction of the Basel II Accord has had a huge impact on financial institutions, allowing them to build credit risk models for three key risk parameters: PD (probability of default), LGD (loss given default) and EAD (exposure at default). Until recently, credit risk research has focused largely on the estimation and validation of the PD parameter, and much less on LGD modeling. In this first large-scale LGD benchmarking study, various regression techniques for modeling and predicting LGD are investigated. These include one-stage models, such as those built by ordinary least squares regression, beta regression, robust regression, ridge regression, regression splines, neural networks, support vector machines and regression trees, as well as two-stage models which combine multiple techniques. A total of 24 techniques are compared using six real-life loss datasets from major international banks. It is found that much of the variance in LGD remains unexplained, as the average prediction performance of the models in terms of R^2 ranges from 4% to 43%. Nonetheless, there is a clear trend that non-linear techniques, and in particular support vector machines and neural networks, perform significantly better than more traditional linear techniques. Also, two-stage models built by a combination of linear and non-linear techniques are shown to have a similarly good predictive power, with the added advantage of having a comprehensible linear model component.
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1. Introduction

With the recent turmoil in credit markets, the topic of credit risk modeling has arguably become more important than ever before. Also, to comply with the Basel II Accord introduced at around the same time, financial institutions have had to invest heavily in the development of improved credit risk models. The Basel II Capital Accord sets out a framework that regulates the minimum amount of capital that financial institutions are required to hold as a safety cushion against unexpected credit-, market- and/or operational losses. More specifically, the accord allows institutions to build credit risk models for three key risk parameters: probability of default (PD), loss given default (LGD) and exposure at default (EAD). From these, the regulatory capital is then derived.

So far, credit risk research has largely focused on the estimation and validation of the PD parameter. On the other hand, the LGD parameter measures the economic loss, expressed as a percentage of the exposure, in case of default. In other words, LGD is the proportion of the remaining loan amount that the bank would not be able to recover. This parameter is a crucial input to the Basel II regulatory capital calculations, as it enters the capital requirement formulas in a linear way (unlike PD, which therefore has less of a direct effect on minimum capital). Hence, any changes in the LGD estimates produced by

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Table 1
Overview of dataset characteristics.

Dataset	Type	Inputs	Total size	Training size	Test size
BANK1	Personal loans	44	47 853	31 905	15 948
BANK2	Mortgage loans	18	119 211	79 479	39 732
BANK3	Mortgage loans	14	3 351	2 232	1 119
BANK4	Revolving credit	12	7 889	5 260	2 629
BANK5	Mortgage loans	35	4 097	2 733	1 364
BANK6	Corporate loans	21	4 276	2 851	1 425

models have a strong bearing on the capital of a financial institution, and thus its long-term strategy as well. It is therefore crucial to have models that estimate LGD as accurately as possible. This is not straightforward, however, as industry models typically show low R^2 values, particularly for consumer lending portfolios. Such models are often built using ordinary least squares regression or regression trees (Bastos, 2009; Bellotti & Crook, 2007; Caselli & Querci, 2009; Gupton & Stein, 2002). Using a set of six real-life default loss datasets, this first large-scale LGD benchmarking study investigates whether or not other approaches can improve the prediction performances of these LGD models.

The remainder of this paper is organized as follows. Section 2 gives an overview of the regression techniques examined, the performance metrics used to evaluate and compare the models, the available datasets and the experimental set-up used in this study. Next, Section 3 reports and discusses the experimental results obtained, and Section 4 concludes the paper.

2. Datasets, techniques and methodology

This section describes the datasets, regression techniques and performance metrics used in this study. We subsequently outline the experimental benchmarking framework used to assess the performances of the various models built on these datasets.

2.1. Dataset characteristics

Table 1 shows the characteristics of six real-life LGD datasets obtained from a series of financial institutions, each of which contains loan-level data about defaulted loans and their resulting losses. The number of dataset entries varies from a few thousands to just under 120,000 observations. The number of available input variables ranges from 12 to 44. The types of loan portfolios included are personal loans, corporate loans, revolving credit and mortgage loans. The empirical distribution of LGD values observed in each of the datasets is displayed in Fig. 1. Note that the LGD distribution in consumer lending often contains spikes around LGD $=$ 0 (in which case there was a full recovery) and/or LGD $=$ 1 (no recovery). Also, some of the datasets include some LGD values that are either negative (e.g., because of penalties paid, gains in collateral sales, etc.) or larger than 1 (e.g., due to additional collection costs incurred), while in other datasets, values outside the unit interval were truncated to 0 or 1 by the banks themselves. Importantly, LGD does not appear to be normally distributed in any of these datasets.

2.2. Regression techniques

The experiments comprise a selection of one-stage and two-stage techniques, listed in Table 2. The one-stage techniques can be divided into linear and non-linear techniques. The linear techniques included in our study model the (original or transformed) dependent variable as a linear function of the independent variables, whereas the non-linear techniques fit a non-linear model to the dataset. Two-stage models are a combination of the aforementioned one-stage models. These either combine the comprehensibility of an OLS model with the added predictive power of a non-linear technique, or use one model to first discriminate between zero- and higher LGDs and a second model to estimate LGD for the subpopulation of non-zero LGDs. Further details are provided in Table 2.

2.3. Performance metrics

Several different performance metrics can be used to evaluate the extent to which the models produce accurate predictions for the dependent variable. Each of the metrics listed in Table 3 has its own method of quantifying the model performance, as specified in the second column of the table. The next two columns show the metric values for the worst (null model) and best possible performances, respectively.[1] The final column indicates whether the metric evaluates calibration or discrimination (Van Gestel & Baesens, 2009). Calibration indicates how close the predicted values are to the observed values, whereas discrimination refers to the ability to provide an ordinal ranking of the dependent variable. A good ranking does not necessarily imply a good calibration. On the other hand, a good calibration always implies good discrimination.

2.4. Experimental set-up

After data preprocessing, the models are built on the training sets and performance metrics are reported for the test sets. Several of the techniques included require setting or tuning parameters, and/or benefit from variable selection; further details of both are provided below, along with the procedure used to assess whether the observed performance differences are statistically significant or not.

[1] Note that the R^2 measure defined here could possibly lie outside the [0,1] interval when applied to non-OLS models. Although alternative generalized goodness-of-fit measures have been put forward for evaluating various non-linear models (see for example the R^2 of Nagelkerke, 1991), the measure defined in Table 3 has the advantage that it is widely used and can be calculated for all techniques.

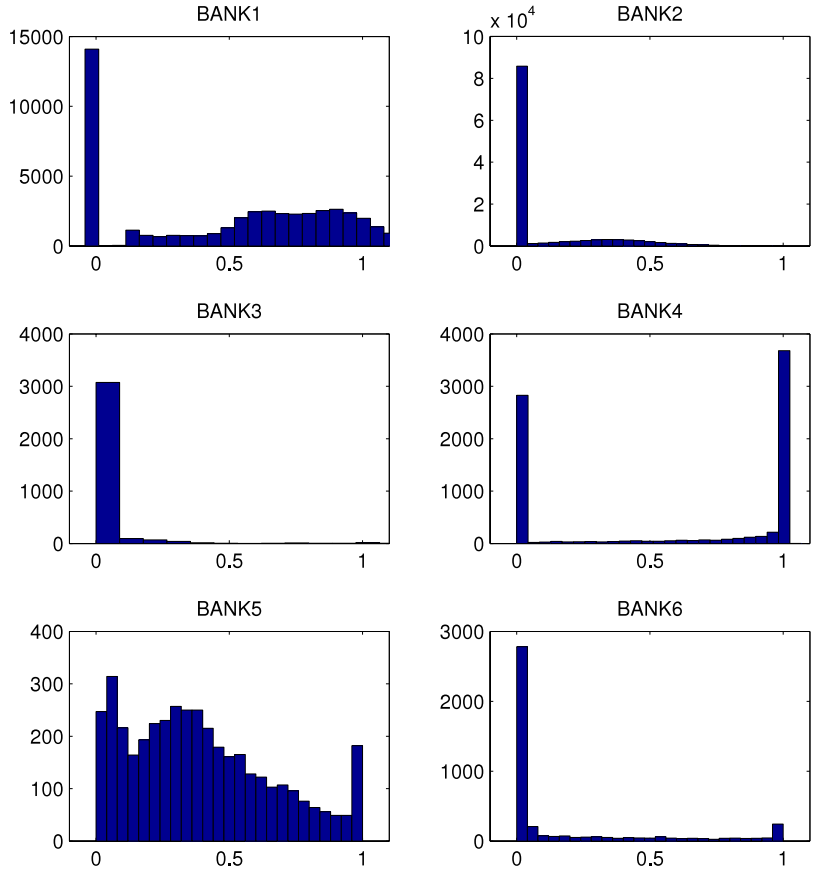


Fig. 1. Empirical LGD distributions for six real-life datasets.

2.4.1. Dataset preprocessing

Each dataset is randomly shuffled and divided into two thirds training set and one third test set. The training set is used to build the models, while the test set is used solely to assess the prediction performances of the models. Where appropriate, continuous independent variables are standardized with the sample mean and standard deviation of the training set; nominal independent variables are dummy encoded and ordinal independent variables are thermo-encoded.

2.4.2. Variable selection

Input selection methods are used to remove irrelevant or redundant independent variables from the datasets, as this may improve the performance of the resulting regression models. More specifically, a stepwise selection procedure is applied in building the linear models, i.e. OLS, B-OLS, BR, BC-OLS, RiR and RoR. For computational efficiency reasons, an R^2-based filter method (Freund & Littell, 2000) is applied prior to building the LSSVM and ANN models. Both RT and MARS already perform variable selection implicitly, so no additional input selection is required here.

2.4.3. Parameter settings and tuning

OLS, B-OLS and BR can be run without the need for any parameter tuning. For RiR, the ridge parameter is tuned by 10-fold cross validation on the training set. The values are varied between 0 and 1 in steps of 0.01, and the mean squared error is used as the selection criterion. For RoR, the commonly used bisquare function is chosen as the objective function, and its parameter k is set to 4.685 times the standard deviation of the residual (Hampel, Ronchetti, Rousseeuw, & Stahel, 1986). The value of the power parameter for the BC-OLS models is varied over a chosen range, i.e. from -3 to 3 in increments of 0.25, and the optimal value is chosen based on a maximum likelihood criterion.

For the RT model, the training set is further split into training and validation subsets. The validation set is used to select the criterion for evaluating candidate splitting rules (i.e. variance reduction or ProbF), the depth of the tree and the threshold p-value for the ProbF criterion. They were all selected based on the mean squared error on the validation set. To run MARS, we set the penalty for adding a hinge function to 2.5 (Hastie, Tibshirani, & Friedman, 2001); the maximum interaction degree is varied from 0 to 5 in steps of 1, and a setting is chosen based on the mean squared error using 10-fold cross validation on the training set.

For LSSVM regression, the radial basis function (RBF) kernel is used because of its good overall performance for LSSVM classifiers (Baesens et al., 2000). Its hyperparameters are again tuned using 10-fold cross validation on the training dataset. A grid search procedure evaluates a large

Table 2
Overview of the included regression techniques and their characteristics.

Technique	Description
Linear	
OLS	Ordinary least squares estimation (e.g. Draper & Smith, 1998) produces a linear regression model that minimizes the sum of squared residuals for the dataset.
B-OLS	Before estimating an OLS model, Beta transformation/OLS (Gupton & Stein, 2002) fits a Beta distribution to the dependent variable (LGD), based on which that variable is transformed to better meet the OLS normality assumption.
BR	Beta regression (Smithson & Verkuilen, 2006) uses maximum likelihood estimation to produce a generalized linear model variant that allows for a dependent variable that is beta-distributed conditional on the input variables.
BC-OLS	Box–Cox transformation/OLS (Box & Cox, 1964) selects an instance of a family of power transformations to improve the normality of the dependent variable.
RiR	Ridge regression (Hoerl & Kennard, 1970) uses an alternative criterion to OLS that is less sensitive to correlated independent variables.
RoR	Robust regression (Holland & Welsch, 1977) uses a criterion that is less sensitive to outliers.
Non-linear	
RT	Regression tree (Breiman, Friedman, Stone, & Olshen, 1984) algorithms produce a decision tree for the dependent variable by recursively partitioning the input space based on a splitting criterion, e.g. a weighted reduction of the within-node variance.
MARS	Multivariate adaptive regression splines (Friedman, 1991) is a stepwise regression algorithm that builds a model from piecewise linear basis function (spline) pairs and allows for pairwise or higher-order products of these functions.
LSSVM	Least squares support vector machine (Suykens, Van Gestel, Van Gestel, De Moor & Vandewalle, 2003; Vapnik, 1995; Wang & Hu, 2005) regression implicitly maps the input space to a kernel-induced high-dimensional feature space in which a linear relationship is fitted.
ANN	Artificial neural networks (Bishop, 1995) produce an output value by feeding inputs through a network whose subsequent nodes apply some chosen activation function to a weighted sum of incoming values. The type of ANN considered in this study is the popular multilayer perceptron (MLP).
Log + (non-)linear: LOG + OLS, B-OLS, BR, BC-OLS, RiR, RoR, RT, MARS, LSSVM or ANN	This class of two-stage (mixture) modeling approaches (Matuszyk, Mues, & Thomas, 2010) uses logistic regression (see e.g. Hosmer & Stanley, 2000) to first estimate the probability of LGD ending up in the peak at 0 (i.e. LGD ≤ 0) or to the right of it (i.e. LGD > 0). A second-stage (non-)linear regression model is built using only the observations for which LGD > 0. An LGD estimate is then produced by weighting the average LGD in the peak and the estimate produced by the second-stage model by their respective probabilities.
Linear + non-linear: OLS + RT, OLS + MARS, OLS + LSSVM, OLS + ANN	The purpose of these two-stage techniques (Van Gestel et al., 2007) is to combine the good comprehensibility of OLS with the predictive power of a non-linear regression technique. In a first stage, a linear model is built with OLS. In a second stage, the residuals of this linear model are estimated using a non-linear regression model. This estimate of the residuals is then added to the OLS estimate to obtain a more accurate prediction for LGD.

Table 3
Performance metrics and their characteristics.

Metric	Description	Worst	Best	Type
RMSE	RMSE (Draper & Smith, 1998) is the square root of the average squared difference between the predicted and observed values.	$+\infty$	0	Calibration
MAE	MAE (Draper & Smith, 1998) is the mean absolute difference between the predicted and observed values.	$+\infty$	0	Calibration
AUC	AUC refers to the area under the ROC curve (Fawcett, 2006), which is used here to plot how good the model is at distinguishing values that are higher than average from those that are lower than average.	0.5	1	Discrimination
AOC	AOC refers to the area above the REC curve (Bi & Bennet, 2003), which plots the percentage of accurate predictions against an increasing error tolerance bound.	$+\infty$	0	Calibration
R^2	The coefficient of determination R^2 (Draper & Smith, 1998) is defined as 1 minus the fraction of the residual sum of squares to the total sum of squares, and thus measures the proportion of the variance that can be explained by the model.	0	1	Calibration
r	Pearson's r (Cohen, Cohen, West, & Aiken, 2002) measures the degree of linear relationship between predictions and observations.	0	1	Discrimination
ρ	Spearman's ρ (Cohen et al., 2002) assesses how well the relationship between the predictions and observations can be described using a monotonic function.	0	1	Discrimination
τ	Kendall's τ (Cohen et al., 2002) measures the degree of correspondence between predictions and observations.	0	1	Discrimination

space of possible hyperparameter combinations so as to find the combination that minimizes the mean squared error. The limits of the grid for the kernel and regularization parameter are set to $\left[0.5\sqrt{n}, 500\sqrt{n}\right]$ and $\left[\frac{0.01}{m}, \frac{1000}{m}\right]$, where n and m denote the numbers of observations and variables, respectively (Van Gestel et al., 2003). On a larger

Table 4
BANK1 model performance results.

Technique	MAE	RMSE	AUC	AOC	R^2	r	ρ	τ
OLS	0.3257	0.3716	0.6570	0.1380	0.0972	0.3112	0.3084	0.2145
B-OLS	0.3474	0.4294	0.6580	0.1843	−0.2060	0.2954	0.2991	0.2071
BR	0.3356	0.3693	0.5690	0.1363	0.0546	0.2601	0.2641	0.1844
BC-OLS	0.3835	0.4579	0.5180	0.2096	−0.3747	0.2403	0.2312	0.1602
RiR	0.3267	0.3723	0.6561	0.1385	0.0933	0.3056	0.3033	0.2106
RoR	0.3262	0.3723	0.6565	0.1385	0.0935	0.3061	0.3034	0.2107
RT	0.3228	0.3732	0.5990	0.1392	0.0892	0.2997	0.2913	0.2095
MARS	0.3214	0.3704	0.6657	0.1372	0.1027	0.3205	0.3122	0.2187
LSSVM	0.3184	0.3669	0.6723	0.1346	0.1194	0.3466	0.3442	0.2444
ANN	0.3118	0.3648	0.6840	0.1331	0.1295	0.3603	0.3559	0.2524
LOG + OLS	0.3202	0.3700	0.6210	0.1366	0.1063	0.3262	0.3143	0.2214
LOG + B-OLS	0.3163	0.3750	0.6020	0.1406	0.1002	0.3166	0.3103	0.2185
LOG + BR	0.3560	0.4142	0.5270	0.1715	0.0782	0.2797	0.2591	0.1794
LOG + BC-OLS	0.4308	0.5090	0.5040	0.2590	−0.6946	0.2125	0.2440	0.1731
LOG + RiR	0.3193	0.3693	0.6655	0.1363	0.1081	0.3289	0.3167	0.2234
LOG + RoR	0.3171	0.3700	0.6554	0.1369	0.1045	0.3264	0.3205	0.2270
LOG + RT	0.3219	0.3693	0.6160	0.1363	0.1081	0.3301	0.3212	0.2263
LOG + MARS	0.3205	0.3689	0.6658	0.1360	0.1099	0.3320	0.3248	0.2286
LOG + LSSVM	0.3191	0.3679	0.6664	0.1353	0.1150	0.3401	0.3336	0.2371
LOG + ANN	0.3174	0.3664	0.6320	0.1342	0.1221	0.3502	0.3406	0.2395
OLS + RT	0.3170	0.3681	0.6730	0.1354	0.1137	0.3382	0.3342	0.2348
OLS + MARS	0.3177	0.3679	0.6799	0.1353	0.1150	0.3394	0.3363	0.2352
OLS + LSSVM	0.3115	0.3631	0.6929	0.1317	0.1379	0.3714	0.3666	0.2596
OLS + ANN	0.3079	0.3633	0.6960	0.1318	0.1367	0.3716	0.3638	0.2581

dataset, this search process can be computationally intensive, and therefore a random sample of 4000 observations is chosen from the complete training set for the purpose of tuning the LSSVM hyperparameters before the final model is run on the full training set.

Finally, in order to train the ANNs, we again split each training set into a training set and a validation set. The validation-set mean squared error is then used to select the target layer activation function (logistic, linear, exponential, reciprocal, square, sine, cosine, tanh or arcTan) and determine the number of hidden neurons (a value from 1 to 20 is considered). The hidden layer activation function is set to logistic.

2.4.4. Model evaluation

The performances of the resulting models are measured on the test set according to the eight performance metrics listed in Table 3. For each dataset, the techniques are ranked from 1 (best) to 24 (worst) based on the resulting values for each of these metrics. Then, the average rank of each technique over all datasets is calculated for each metric. To further summarize the results, an overall average ranking of techniques over the datasets and over all metrics is also produced.

2.4.5. Statistical comparison

The model performances are compared statistically using Friedman's test (Friedman, 1940) and the post-hoc multiple testing procedure of Hommel (1988), as suggested in the literature (Demsar, 2006; Garcia & Herrera, 2008). Friedman's test is performed to test the null hypothesis that all regression techniques perform alike, based on their rankings for a chosen performance metric. We then use Hommel's method to compare each regression technique against the best performing one and report significant rank differences. All statistical tests are based on the 95% confidence level.

3. Results and discussion

Tables 4–9 contain the performance results obtained for each technique on the six respective datasets. The best performing model according to each metric is underlined. The Friedman test results for the respective performance measures all indicate that the observed differences in performance are extremely unlikely to be due to random chance (e.g., for R^2, the p-value is 1.06E−08). Fig. 2 displays a series of box plots for the observed distributions of performance values for the metrics AUC, R^2, r, ρ and τ. Similar trends can be observed across all metrics. Note that differences in the type of portfolio, number of observations and available independent variables are the probable causes of the observed variability in the actual performance levels between the six different datasets.

Although all of the performance metrics listed above are useful measures in their own right, it is common to use the coefficient of determination R^2 to compare the model performances, since the R^2 value measures the calibration and can be compared meaningfully across different datasets. As is shown in Fig. 2, the average R^2 of the models varies from about 4% to 43%. In other words, the variance in LGD that can be explained by the independent variables is consistently below 50%, implying that most of the variance cannot be explained, even with the best models. Note that although R^2 is usually a number on the scale of 0–1, R^2 can yield negative values for non-OLS models when the model predictions are worse than always using the mean from the training set as the prediction.

Table 10 shows the average rankings of the techniques over the six datasets according to each performance metric. In addition, their overall average rank over the six datasets and the eight performance metrics is included in the last column. The techniques are ordered according to their overall average ranking. The best performing technique for each metric is again underlined, and techniques

Table 5
BANK2 model performance results.

Technique	MAE	RMSE	AUC	AOC	R^2	r	ρ	τ
OLS	0.1187	0.1613	0.8100	0.0259	0.2353	0.4851	0.4890	0.3823
B-OLS	0.1058	0.1621	0.8000	0.0262	0.2273	0.4768	0.4967	0.3881
BR	0.1020	0.1661	0.7300	0.0275	0.2120	0.4635	0.4857	0.3861
BC-OLS	0.1056	0.1623	0.7450	0.0262	0.2226	0.4718	0.4990	0.3900
RiR	0.1187	0.1606	0.8074	0.0258	0.2415	0.4915	0.4855	0.3792
RoR	0.1075	0.1663	0.8063	0.0277	0.1866	0.4770	0.4824	0.3751
RT	0.0978	0.1499	0.7710	0.0224	0.3390	0.5823	0.5452	0.4357
MARS	0.1068	0.1531	0.8397	0.0234	0.3113	0.5579	0.5321	0.4168
LSSVM	0.1047	0.1518	0.8365	0.0230	0.3229	0.5690	0.5301	0.4160
ANN	0.0956	0.1472	0.8530	0.0216	0.3632	0.6029	0.5549	0.4366
LOG + OLS	0.1060	0.1622	0.7590	0.0255	0.2268	0.4838	0.5206	0.4084
LOG + B-OLS	0.1040	0.1567	0.8320	0.0245	0.2779	0.5286	0.5202	0.4083
LOG + BR	0.1015	0.1688	0.7250	0.0285	0.2024	0.4529	0.4732	0.3876
LOG + BC-OLS	0.1034	0.1655	0.7320	0.0273	0.2124	0.4628	0.4870	0.3820
LOG + RiR	0.1049	0.1554	0.8312	0.0240	0.2901	0.5386	0.5209	0.4091
LOG + RoR	0.1043	0.1558	0.8307	0.0242	0.2859	0.5350	0.5200	0.4084
LOG + RT	0.1041	0.1538	0.8360	0.0236	0.3049	0.5545	0.5254	0.4126
LOG + MARS	0.1031	0.1537	0.8355	0.0236	0.3059	0.5531	0.5268	0.4149
LOG + LSSVM	0.1031	0.1530	0.8334	0.0234	0.3121	0.5587	0.5243	0.4128
LOG + ANN	0.1011	0.1531	0.8430	0.0234	0.3109	0.5585	0.5380	0.4240
OLS + RT	0.1015	0.1506	0.8410	0.0227	0.3331	0.5786	0.5344	0.4188
OLS + MARS	0.1081	0.1526	0.8379	0.0233	0.3150	0.5615	0.5300	0.4156
OLS + LSSVM	0.1029	0.1520	0.8428	0.0230	0.3208	0.5665	0.5398	0.4241
OLS + ANN	0.0999	0.1474	0.8560	0.0217	0.3612	0.6010	0.5585	0.4398

Table 6
BANK3 model performance results.

Technique	MAE	RMSE	AUC	AOC	R^2	r	ρ	τ
OLS	0.0549	0.1411	0.6460	0.0178	0.0124	0.1168	0.0965	0.0718
B-OLS	0.0348	0.1449	0.6610	0.0188	−0.0419	0.0767	0.1754	0.1361
BR	0.0883	0.1315	0.6530	0.0169	−0.1128	0.1567	0.1719	0.1323
BC-OLS	0.0340	0.1456	0.6380	0.0190	−0.0529	0.1373	0.2312	0.1765
RiR	0.0550	0.1405	0.6499	0.0177	0.0210	0.1460	0.1270	0.0936
RoR	0.0347	0.1453	0.6438	0.0189	−0.0464	0.1733	0.1991	0.1501
RT	0.0482	0.1311	0.6990	0.0154	0.1477	0.3869	0.2007	0.1673
MARS	0.0478	0.1229	0.7345	0.0131	0.2506	0.5016	0.1344	0.0974
LSSVM	0.0473	0.1270	0.7441	0.0140	0.1998	0.4526	0.2085	0.1520
ANN	0.0458	0.1318	0.6000	0.0152	0.1386	0.3776	0.1482	0.1105
LOG + OLS	0.0553	0.1417	0.6010	0.0179	0.0043	0.0759	0.0701	0.0510
LOG + B-OLS	0.0392	0.1429	0.6330	0.0182	−0.0127	0.1214	0.1252	0.0923
LOG + BR	0.0569	0.1417	0.5790	0.0180	0.0043	0.0742	0.1710	0.1265
LOG + BC-OLS	0.0349	0.1448	0.6330	0.0188	−0.0395	0.1665	0.1918	0.1426
LOG + RiR	0.0545	0.1408	0.6404	0.0177	0.0169	0.1319	0.1511	0.1094
LOG + RoR	0.0366	0.1440	0.6510	0.0185	−0.0277	0.1510	0.2057	0.1504
LOG + RT	0.0434	0.1297	0.7210	0.0146	0.1663	0.4553	0.1571	0.1170
LOG + MARS	0.0467	0.1264	0.7365	0.0139	0.2082	0.4884	0.1381	0.0998
LOG + LSSVM	0.0460	0.1312	0.7485	0.0151	0.1471	0.4152	0.2272	0.1676
LOG + ANN	0.0452	0.1219	0.6190	0.0133	0.2634	0.5381	0.1671	0.1242
OLS + RT	0.0540	0.1372	0.7050	0.0168	0.0660	0.2578	0.1748	0.1285
OLS + MARS	0.0471	0.1229	0.7189	0.0131	0.2512	0.5018	0.1231	0.0879
OLS + LSSVM	0.0483	0.1258	0.7416	0.0137	0.2148	0.4648	0.1869	0.1354
OLS + ANN	0.0570	0.1388	0.6730	0.0171	0.0442	0.2605	0.1369	0.1005

which perform significantly worse than this best technique according to Hommel's procedure are displayed in italics. It can be observed that the same techniques, LSSVM and ANN, are consistently ranked in the top two, regardless of the metric.

The pure linear models built by OLS, RiR and RoR do not seem to show any consistent differences in performance between one another. Although RiR is ranked somewhat higher overall, it most often leads to a model performance identical to that of OLS. We suspect that any potential benefits of RiR could be limited in this particular setting because the chosen variable selection methods eliminate highly correlated variables a priori. On all datasets, RoR produces models that either perform slightly worse than the OLS models or show similar performances. Hence, RoR's ability to reduce the impact of outliers does not result in any actual performance improvement on our real-life datasets.

The linear models that incorporate some form of transformation to the dependent variable (i.e. B-OLS, BR, BC-OLS) are shown to perform consistently worse than OLS, despite the fact that these approaches are specifically designed to cope with a violation of the OLS normality assumption. This suggests that they too have trouble

Table 7
BANK4 model performance results.

Technique	MAE	RMSE	AUC	AOC	R^2	r	ρ	τ
OLS	0.2712	0.3479	0.8520	0.1208	0.4412	0.6643	0.5835	0.4331
B-OLS	0.2214	0.3743	0.8500	0.1396	0.3530	0.6510	0.5822	0.4321
BR	0.3208	0.3777	0.8480	0.1425	0.3405	0.6527	0.5908	0.4452
BC-OLS	0.3185	0.4292	0.6750	0.1839	0.1478	0.5726	0.5820	0.4316
RiR	0.2707	0.3473	0.8541	0.1204	0.4429	0.6657	0.5972	0.4495
RoR	0.2576	0.3607	0.8483	0.1299	0.3992	0.6527	0.5857	0.4402
RT	0.2476	0.3362	0.8480	0.1128	0.4782	0.6916	0.5919	0.4762
MARS	0.2617	0.3361	0.8636	0.1128	0.4783	0.6917	0.6162	0.4631
LSSVM	0.2428	0.3315	0.8655	0.1097	0.4924	0.7017	0.6203	0.4692
ANN	0.2393	0.3299	0.8670	0.1086	0.4974	0.7053	0.6109	0.4555
LOG + OLS	0.2577	0.3465	0.8520	0.1199	0.4455	0.6678	0.5840	0.4338
LOG + B-OLS	0.2399	0.3551	0.8500	0.1259	0.4176	0.6651	0.5801	0.4301
LOG + BR	0.2738	0.3560	0.8520	0.1265	0.4147	0.6680	0.5868	0.4342
LOG + BC-OLS	0.2502	0.3489	0.8510	0.1215	0.4379	0.6659	0.5819	0.4322
LOG + RiR	0.2538	0.3432	0.8572	0.1176	0.4559	0.6755	0.6026	0.4543
LOG + RoR	0.2354	0.3521	0.8534	0.1238	0.4275	0.6728	0.5960	0.4477
LOG + RT	0.2679	0.3621	0.8570	0.1309	0.3945	0.6656	0.5899	0.4364
LOG + MARS	0.2536	0.3433	0.8572	0.1177	0.4558	0.6754	0.6027	0.4544
LOG + LSSVM	0.2534	0.3425	0.8590	0.1172	0.4581	0.6771	0.6024	0.4541
LOG + ANN	0.2558	0.3457	0.8540	0.1184	0.4480	0.6698	0.5852	0.4348
OLS + RT	0.2628	0.3425	0.8590	0.1171	0.4582	0.6776	0.6017	0.4498
OLS + MARS	0.2617	0.3362	0.8620	0.1128	0.4781	0.6915	0.6117	0.4582
OLS + LSSVM	0.2439	0.3322	0.8656	0.1102	0.4904	0.7003	0.6211	0.4698
OLS + ANN	0.2404	0.3300	0.8710	0.1087	0.4971	0.7053	0.6195	0.4635

Table 8
BANK5 model performance results.

Technique	MAE	RMSE	AUC	AOC	R^2	r	ρ	τ
OLS	0.1875	0.2375	0.7480	0.0555	0.2218	0.4740	0.5192	0.3651
B-OLS	0.1861	0.2368	0.7410	0.0561	0.2263	0.5073	0.5168	0.3636
BR	0.1957	0.2402	0.7240	0.0575	0.2038	0.4557	0.4811	0.3359
BC-OLS	0.1848	0.2373	0.7390	0.0560	0.2228	0.5014	0.5155	0.3632
RiR	0.1864	0.2373	0.7467	0.0555	0.2233	0.4775	0.5238	0.3704
RoR	0.1892	0.2430	0.7406	0.0579	0.1852	0.4543	0.5121	0.3612
RT	0.1851	0.2324	0.7370	0.0538	0.2546	0.5056	0.4957	0.3888
MARS	0.1733	0.2222	0.7709	0.0488	0.3187	0.5666	0.5565	0.3980
LSSVM	0.1707	0.2198	0.7847	0.0479	0.3331	0.5794	0.5801	0.4167
ANN	0.1678	0.2173	0.7830	0.0470	0.3486	0.5964	0.5765	0.4148
LOG + OLS	0.1851	0.2336	0.7500	0.0542	0.2468	0.4975	0.5246	0.3704
LOG + B-OLS	0.1852	0.2347	0.7480	0.0548	0.2397	0.5117	0.5192	0.3658
LOG + BR	0.1939	0.2395	0.7250	0.0572	0.2083	0.4568	0.4820	0.3364
LOG + BC-OLS	0.1833	0.2349	0.7470	0.0549	0.2388	0.5099	0.5238	0.3699
LOG + RiR	0.1854	0.2347	0.7492	0.0547	0.2400	0.4922	0.5274	0.3730
LOG + RoR	0.1877	0.2390	0.7451	0.0567	0.2118	0.4744	0.5190	0.3665
LOG + RT	0.1846	0.2344	0.7380	0.0547	0.2420	0.5000	0.4903	0.3445
LOG + MARS	0.1738	0.2217	0.7726	0.0486	0.3215	0.5687	0.5597	0.3985
LOG + LSSVM	0.1708	0.2197	0.7835	0.0479	0.3340	0.5797	0.5795	0.4163
LOG + ANN	0.1689	0.2188	0.7810	0.0476	0.3396	0.5845	0.5737	0.4135
OLS + RT	0.1779	0.2320	0.7660	0.0530	0.2572	0.5357	0.5554	0.3963
OLS + MARS	0.1713	0.2215	0.7740	0.0484	0.3231	0.5769	0.5707	0.4082
OLS + LSSVM	0.1695	0.2216	0.7882	0.0485	0.3223	0.5755	0.5933	0.4279
OLS + ANN	0.1747	0.2277	0.7730	0.0510	0.2844	0.5567	0.5706	0.4086

dealing with the pronounced point densities observed in LGD datasets, while they could be less efficient than OLS or introduce model bias if a transformation is performed prior to OLS estimation (as is the case for B-OLS and BC-OLS).

Perhaps the most striking result is that, in contrast to prior benchmarking studies on classification models for PD (Baesens et al., 2003), non-linear models such as LSSVM and ANN significantly outperform most of the linear models in the prediction of LGD. This implies that the relationship between LGD and the independent variables in the datasets is non-linear (as is most apparent on dataset BANK3, see Table 6). Also, LSSVM and ANN generally perform better than RT or MARS. However, LSSVM and ANN result in black-box models while RT and MARS have the ability to produce comprehensible white-box models. To circumvent this disadvantage, one could try to obtain an interpretation of a well-performing black box model by applying rule extraction techniques (Martens, Baesens & Van Gestel, 2009; Martens, Baesens, Van Gestel & Vanthienen, 2007).

The performance evaluation of the class of two-stage models in which a logistic regression model is combined with a second-stage (linear or non-linear) model (LOG+), is less straightforward. Although there appears to be a

Table 9
BANK6 model performance results.

Technique	MAE	RMSE	AUC	AOC	R^2	r	ρ	τ
OLS	0.2085	0.2874	0.7180	0.0822	0.1197	0.3502	0.3032	0.2071
B-OLS	0.1783	0.3055	0.7120	0.0933	0.0933	0.3054	0.3112	0.2138
BR	0.2612	0.3019	0.7090	0.0909	0.1029	0.3209	0.3138	0.2151
BC-OLS	0.1824	0.3149	0.7100	0.0988	0.0815	0.2855	0.3139	0.2172
RiR	0.2086	0.2868	0.7200	0.0818	0.1231	0.3544	0.3045	0.2076
RoR	0.2087	0.2875	0.7180	0.0822	0.1189	0.3493	0.3030	0.2070
RT	0.2061	0.2885	0.7040	0.0829	0.1129	0.3390	0.3180	0.2482
MARS	0.2057	0.2856	0.7184	0.0811	0.1302	0.3615	0.3131	0.2251
LSSVM	0.2031	0.2812	0.7360	0.0787	0.1570	0.3964	0.3207	0.2190
ANN	0.2004	0.2860	0.7210	0.0815	0.1281	0.3619	0.2893	0.2000
LOG + OLS	0.2086	0.2876	0.7180	0.0824	0.1182	0.3479	0.3012	0.2060
LOG + B-OLS	0.1899	0.2964	0.7070	0.0875	0.0635	0.3225	0.2913	0.2000
LOG + BR	0.2875	0.3204	0.7070	0.1024	−0.0946	0.3346	0.2806	0.1918
LOG + BC-OLS	0.1863	0.3055	0.7120	0.0930	0.0963	0.3103	0.3050	0.2118
LOG + RiR	0.2062	0.2933	0.7128	0.0856	0.0831	0.3409	0.3150	0.2176
LOG + RoR	0.2060	0.2937	0.7118	0.0858	0.0806	0.3391	0.3162	0.2191
LOG + RT	0.2052	0.2890	0.6880	0.0832	0.1100	0.3348	0.3179	0.2219
LOG + MARS	0.2058	0.2934	0.6942	0.0857	0.0820	0.3285	0.2953	0.2121
LOG + LSSVM	0.2024	0.2887	0.7191	0.0829	0.1116	0.3652	0.3159	0.2190
LOG + ANN	0.2038	0.2854	0.7290	0.0811	0.1319	0.3689	0.3243	0.2216
OLS + RT	0.2066	0.2866	0.7190	0.0817	0.1244	0.3623	0.3067	0.2100
OLS + MARS	0.2068	0.2861	0.7237	0.0815	0.1271	0.3634	0.3081	0.2102
OLS + LSSVM	0.2087	0.2875	0.718	0.0822	0.1189	0.3493	0.3030	0.2070
OLS + ANN	0.2085	0.2874	0.7190	0.0822	0.1200	0.3498	0.3049	0.2086

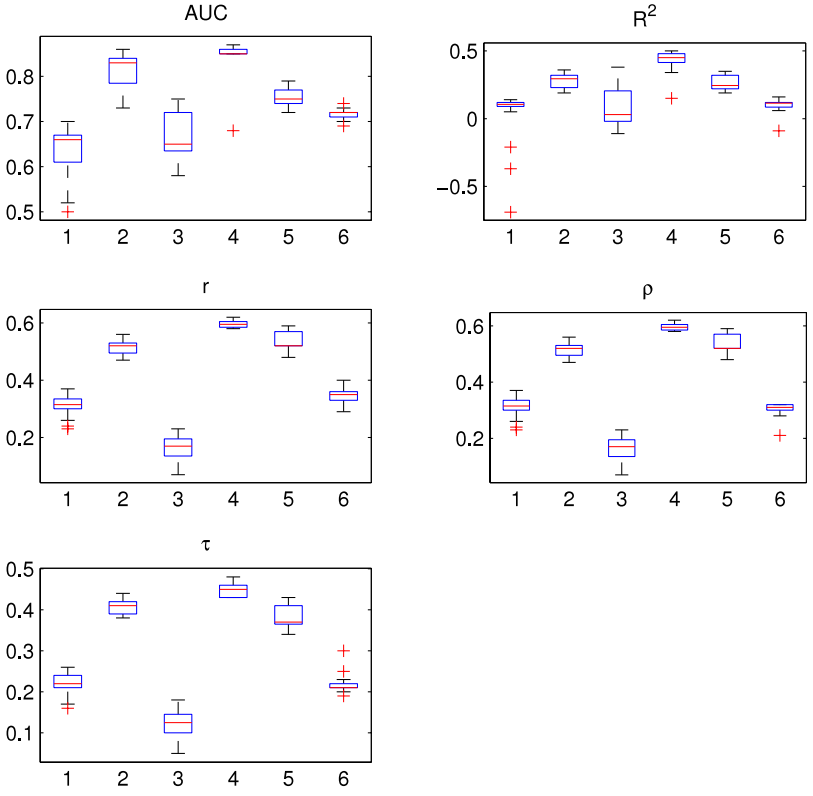


Fig. 2. Variability in the LGD model performances observed for the six datasets.

weak trend that logistic regression combined with a linear model tends to improve the performance of the latter, it appears that logistic regression combined with a non-linear model slightly reduces the strong performance of the latter. Because the LGD distributions from BANK4, BANK5 and BANK6 also show a peak at LGD $= 1$, the performances of these models may be able to be improved by altering the technique slightly. Replacing the (binary) logistic regression component with an ordinal logistic regression model distinguishing between 3 classes (LGD ≤ 0,

Table 10
Mean performance ranks of the techniques over the six datasets and the overall average rank (OAR) over all metrics.

Rank	Technique	RMSE	MAE	AUC	AOC	R^2	r	ρ	τ	OAR
1	LSSVM	4.0	9.0	3.8	4.0	4.5	4.2	3.5	4.8	4.7
2	ANN	3.5	3.7	6.3	3.2	3.8	3.8	8.8	8.9	5.3
3	OLS + LSSVM	5.3	9.6	4.1	5.0	5.8	5.8	5.5	6.2	5.9
4	LOG + ANN	5.1	7.0	10.0	5.4	5.7	5.2	7.8	8.3	6.8
5	OLS + ANN	5.7	9.3	4.6	7.8	6.8	5.8	7.7	7.8	6.9
6	LOG + LSSVM	8.2	8.4	5.9	8.3	8.8	6.2	6.7	6.6	7.4
7	OLS + MARS	5.4	13.4	5.5	5.1	5.9	5.3	10.0	11.0	7.7
8	OLS + RT	8.5	12.2	7.2	6.2	8.5	7.8	9.3	10.3	8.8
9	MARS	7.2	13.6	7.2	7.0	7.5	7.8	10.5	9.5	8.8
10	LOG + MARS	10.0	10.6	10.4	10.1	11.2	10.3	11.8	11.0	10.7
11	RT	10.1	11.4	*19.1*	10.4	10.5	11.2	10.8	6.5	11.2
12	LOG + RiR	13.2	14.7	12.6	13.0	15.1	14.2	11.0	11.8	13.2
13	LOG + RT	13.2	12.5	14.8	13.2	13.6	13.0	12.7	13.2	13.3
14	LOG + RoR	*17.3*	10.0	15.0	*17.3*	*18.2*	14.8	10.8	10.7	14.3
15	RiR	14.5	*20.2*	12.8	14.6	*15.8*	*15.8*	*16.8*	*17.4*	16.0
16	LOG + OLS	14.5	*16.8*	*16.3*	13.8	*15.7*	*16.5*	*16.8*	*17.5*	16.0
17	LOG + B-OLS	*17.6*	7.7	*17.3*	*17.7*	*18.5*	*16.7*	*18.6*	*19.3*	16.7
18	B-OLS	*20.4*	10.3	15.5	*21.2*	*15.8*	*20.2*	*16.0*	16.5	17.0
19	LOG + BC-OLS	*19.9*	10.3	*19.0*	*19.9*	*14.7*	*18.5*	*16.8*	*17.0*	17.0
20	OLS	*15.4*	*19.7*	13.8	*15.3*	*17.0*	*17.3*	*18.4*	*19.5*	17.1
21	RoR	*19.7*	*16.4*	*16.3*	*19.5*	*18.8*	*17.7*	*17.8*	14.0	17.5
22	BC-OLS	*22.1*	13.2	*20.7*	*22.1*	*16.7*	*20.8*	15.2	15.8	18.3
23	BR	*18.2*	*20.3*	*20.3*	*18.5*	*19.8*	*20.8*	*16.7*	16.7	18.9
24	LOG + BR	*21.3*	*19.8*	*21.6*	*21.3*	*21.4*	*20.5*	*20.0*	*19.7*	20.7

$0 < \text{LGD} < 1, \text{LGD} \geq 1$) and then using a second-stage model for $0 < \text{LGD} < 1$ could perhaps account for the presence of the two peaks better.

In contrast to the previous class of two-stage models, a clear trend can be observed for the combination of a linear and a non-linear model (OLS+). By estimating the error residual of an OLS model using a non-linear technique, the prediction performance tends to increase to somewhere very near the level of the corresponding one-stage non-linear technique. The attraction of these two-stage models is that they have the advantage of combining the good prediction performance of non-linear regression with the comprehensibility of a linear regression component. Note that this modeling method has also been applied successfully in a PD modeling context (Van Gestel et al., 2006, 2005, 2007).

4. Conclusions

In this first large-scale LGD benchmarking study, 24 regression techniques were evaluated on six real-life datasets obtained from major international banking institutions. The average performances of the models in terms of R^2 ranged from 4% to 43%, showing that several of the resulting models have a limited explanatory power. Nonetheless, a clear trend can be seen that non-linear techniques, and support vector machines and artificial neural networks in particular, yield significantly better model performances than those of more traditional linear techniques. This suggests the presence of non-linear relationships between the independent variables and LGD, contrary to previous benchmarking studies on PD modeling, where the difference between linear and non-linear techniques was not that explicit. Therefore, the study has clearly demonstrated the potential of applying non-linear techniques to LGD modeling, possibly combined with a linear model component in a two-stage setting so as to improve the comprehensibility of the resulting models.

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